When Is More Computation Worth It? Statistical Foundations for Aggregating Dependent AI Reasoning Traces

Brief topic description for Program Officer discussion · FY27 ONR Young Investigator Program (N0001426SF004)

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Problem. Large language models (LLMs) answer hard questions better when they spend more computation at inference time: they sample many reasoning traces, score intermediate steps with learned verifiers, and keep the majority answer or the best-scoring trace. These rules are heuristics. They treat samples as independent, yet traces from one model share prefixes, retrieved documents, and systematic misreadings, and the verifier's mistakes are often correlated with the generator's. Suppose an onboard assistant estimates whether a ship can complete a transit on its remaining fuel, and 15 of 20 sampled traces reuse the same misread consumption rate. The vote is confidently wrong, and twenty times the computation bought little information. Current voting and stopping rules largely ignore this dependence, so they cannot say how many samples a question needs or when more computation cannot help.

Central hypothesis. Treating sampled traces and verifier scores as dependent, noisy measurements of the latent validity of shared reasoning steps will (i) make the *effective number of independent traces* estimable from few labels; (ii) yield sampling, checking, and stopping rules that reach a target error rate with less computation than voting or fixed-budget best-of-N selection; and (iii) identify when further samples cannot reduce error and an outside check is required.

Research thrusts.

- **A measurement model for dependent reasoning.** Represent a set of traces as paths through shared intermediate claims, each with a latent validity, and model verifier scores as biased, noisy observations whose errors may correlate with the generator's. Study what is identifiable from a small labeled calibration set and many unlabeled traces, estimate the effective sample size, and bound what cannot be identified.
- **Sequential allocation and stopping.** Cast drawing another trace, verifying a step, retrieving a source, and stopping or abstaining as a sequential decision with costs in tokens, latency, and energy. Derive value-of-information rules, determine which sequential-testing and distribution-free risk-control guarantees survive dependent samples and distribution shift, and characterize computation–accuracy frontiers.
- **Integrating heterogeneous checks.** Combine process verifiers, agreement among retrieved sources, and tool-executed checks, whose errors are correlated in different ways. Estimate their dependence, favor checks that add independent information, and identify regimes where more of the same computation raises cost but not accuracy.

What is new. Inference-time scaling is studied mostly empirically, assuming independent votes or fixed best-of-N budgets. We model dependence among a system's own traces and checks and turn it into allocation and stopping rules. Our verifiers (FG-PRM) and efficient-reasoning methods (HiPRAG, AALC) supply the instruments; modeling their dependence is new.

Evaluation. Step-level error identification (ProcessBench), mathematical reasoning (MATH), and multi-hop question answering (MuSiQue), plus controlled shared-error injection that plants a common faulty premise in a chosen fraction of traces so that the true dependence is known. Measures: accuracy–computation curves, selective error at fixed coverage, calibration, and error in the estimated effective sample size. Baselines: majority vote (self-consistency), best-of-N with a process reward model, and adaptive stopping rules that assume independent samples.

Naval relevance. Shipboard and edge systems run AI within fixed power, latency, and bandwidth budgets, often without reliable reach-back. A principled rule tells such a system when one more local sample is worth its cost, when to request an outside check, and when to abstain. The topic answers the program's call to replace heuristics with rigorous mathematical answers, and its interests in balancing limited computation against accuracy and in decision making under uncertainty.

PI foundation. FG-PRM, process reward models trained on synthetic hallucinations (EMNLP Findings 2025); HiPRAG, hierarchical process rewards for efficient agentic retrieval (ICLR 2026); AALC, accuracy–length control for efficient reasoning (arXiv 2025); Search Wisely, reducing sub-optimal agentic search (EMNLP 2025); PRD, aggregating LLM evaluators (TMLR 2024). NSF CAREER award (2024); Amazon Research Award (2023) for process-guided fine-tuning. As a postdoc, the PI led system integration for the RESIN team in DARPA KAIROS (RESIN-11, NAACL 2022 Demo, first author).